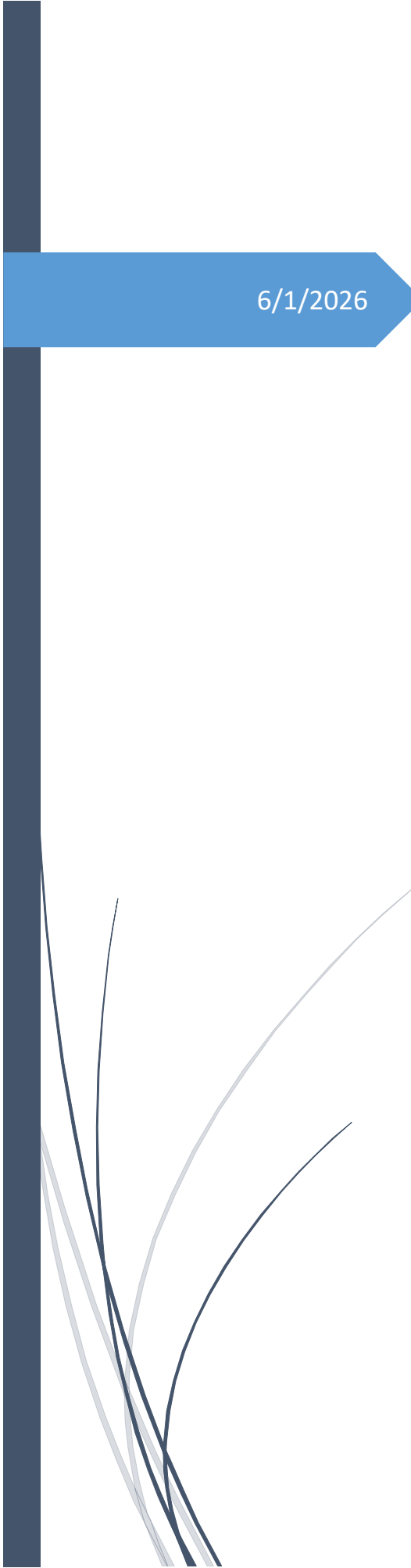




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Gentle Dose

- **Subject:** SPM
- **Class:** BS (SE) 6th
- **Submitted To:** Sir Hafiz M Anas Ali
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Gentle Dose

1. PROBLEM STATEMENT

Many elderly people and patients with chronic illnesses struggle to remember their medications and medical appointments. Missing medicines can lead to serious health complications and reduced treatment effectiveness. Traditional methods such as handwritten notes or relying on memory are often unreliable.

Existing reminder applications may contain complex interfaces, unnecessary features, or limited caregiver support, making them difficult for some users to adopt. Therefore, there is a need for a simple and accessible solution that helps patients manage their daily healthcare activities while allowing caregivers to stay informed.

Gentle Dose addresses this problem by providing an easy-to-use mobile application that sends reminders, tracks medication progress, manages appointments, and supports caregiver involvement.

2. REQUIREMENT ANALYSIS

The proposed system will provide the following functionalities:

Functional Requirements

1. The system shall allow users to register and log in securely.
2. The system shall allow patients to add, update, and delete medications.
3. The system shall allow users to schedule medical appointments.
4. The system shall generate reminders for medications and appointments.
5. The system shall allow patients to add caregivers.
6. The system shall allow caregivers to view patient progress.
7. The system should provide a calendar view for medications and appointments.
8. The system shall track complete and missed medications.
9. The system shall allow users to manage their profile information.
10. The system shall store medication and appointment data for future access.

Non-Functional Requirements

1. The application should be easy to use and understand.
2. The system should provide fast response times.
3. The application should maintain data accuracy and reliability.
4. The user interface should be responsive on different mobile devices.
5. The system should handle invalid user input properly.
6. The application should provide secure user authentication.
7. The application should support offline access to saved medication and appointment records.

Project Management Approach

The project was managed using Agile Scrum methodology. Development activities were divided into multiple sprints. Jira was used for project planning, issue tracking, sprint management, backlog management, and progress monitoring. Confluence was used for project documentation, meeting notes, and project planning records.

3. TESTING AND RESULTS

1. Testing

Several testing activities were performed during the development of Gentle Dose to ensure that the application worked correctly and met project requirements.

2. Functional Testing

The following features were tested:

- User Registration
- User Login
- Medication Management
- Appointment Scheduling
- Caregiver Management
- Notification System
- Calendar View
- Profile Management

3. User Interface Testing

The application interface was tested on different screen sizes to ensure responsiveness and usability.

4. Database Testing

SQLite and Firebase integration were tested to verify that data was stored and retrieved correctly.

5. Notification Testing

Medication reminders and appointment notifications were tested to ensure that alerts were delivered at the correct time.

6. Results

The testing results showed that all major functionalities worked successfully. Users were able to manage medications, schedule appointments, receive reminders, and track their progress without major issues. The caregiver management feature also worked as expected.

Minor issues identified during testing were fixed before project completion. Overall, the application met the defined project requirements and achieved the planned objectives.

4. CONCLUSION

Gentle Dose is a useful healthcare mobile application that helps patients manage medications and appointments more effectively. The project successfully achieved its objectives by providing reminder services, caregiver support, progress tracking, and appointment management features.

The use of Agile project management practices, Jira for issue tracking, and Confluence for documentation helped the team organize development activities efficiently. The final application provides a practical solution for reducing missed medications and improving daily healthcare management.

In the future, additional features such as multilingual support, doctor integration, medicine refill reminders, and advanced health monitoring can further enhance the application.